

# Population structure and maternal genetic diversity in Friuli Venezia Giulia, northeastern Italy

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## Geographic context

Historically a frontier land of mountains and sea, where the Balkans and the Alps meet the Mediterranean, Friuli Venezia Giulia has long served as a major corridor for human migrations, whose multiple waves of movement and interaction have shaped its exceptional cultural and linguistic diversity and complex demographic history. Several distinct dialects are spoken, and the traditions and surnames of several villages link them to ethnic groups from farther away than Friuli Venezia Giulia or Italy<sup>1</sup>. These peculiar features inspired this study, which focuses on the population genomics variation in this region using a diachronic approach.

## Results

Human mtDNA haplotypes from present-day Friuli Venezia Giulia and the archaeological site of San Pietro al Natisone revealed a high level of mtDNA variation, evidenced by 171 haplogroups and sub-haplogroups, which have been grouped into 34 macro-haplogroups (**Fig. A**).

Macro-haplogroups distributions differ across Friuli Venezia Giulia localities. Erto, Sauris, and Illegio appear more isolated, whereas Resia, San Martino del Carso and Clauzetto show similar macro-haplogroup distributions. In this context, the archaeological site of San Pietro al Natisone is closer to Resia (**Fig. B**). These observations supported further demographic analyses using Bayesian Skyline Plots (BSPs)<sup>2</sup> for each municipality. BSP analyses suggest a decline in effective population size ( $N_e$ ) during the last ~1,500 years. However, when each haplotype was counted only once, this signal disappeared, indicating that the inferred contraction was driven by the overrepresentation of a few maternal lineages. These results support increased local isolation and reduced gene flow rather than a true demographic collapse (**Fig. C**). In this light, after testing different models, Approximate Bayesian Computation (ABC)<sup>3</sup> analyses support the presence of significant mitochondrial genetic structure across the region nowadays, indicating that isolation has contributed to shaping mitochondrial diversity (**Fig. D**).

## Future perspectives

To further explore these peculiarities, we obtained whole-genome shotgun data of 12 individuals from the ancient DNA collection (average coverage ~2X). This dataset will enable allele- and haplotype-based analyses<sup>4, 5, 6</sup> and can be used to verify whether mitochondrial patterns reflect only localized maternal processes, and to clarify how isolation and migration shaped genetic variation in Friuli Venezia Giulia over time.

### Dataset

402 modern mtDNA  
25 ancient mtDNA

### Aim



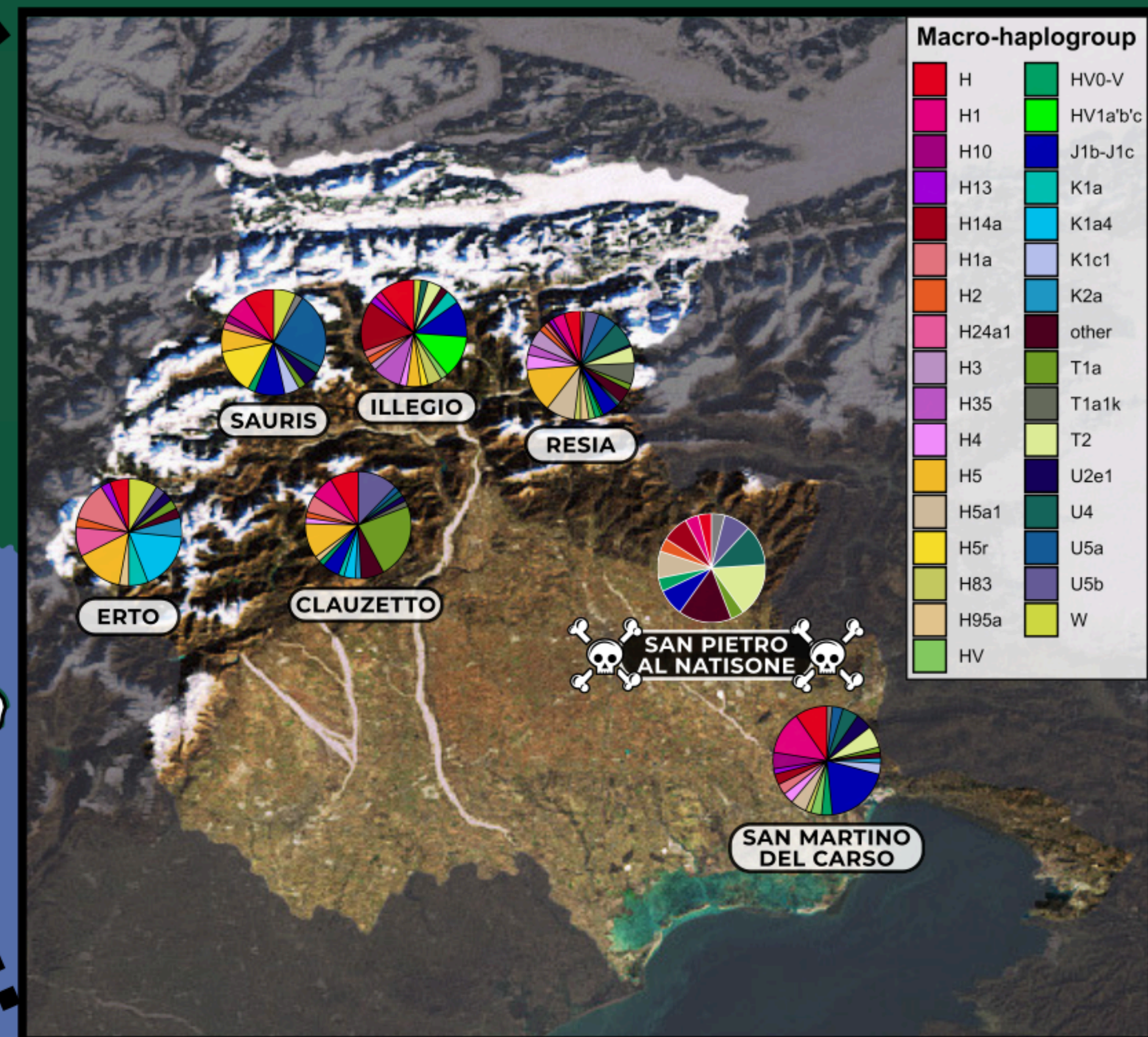
Describe mitochondrial variation and possible genetic structure



Investigate mitochondrial gene pool evolution through time



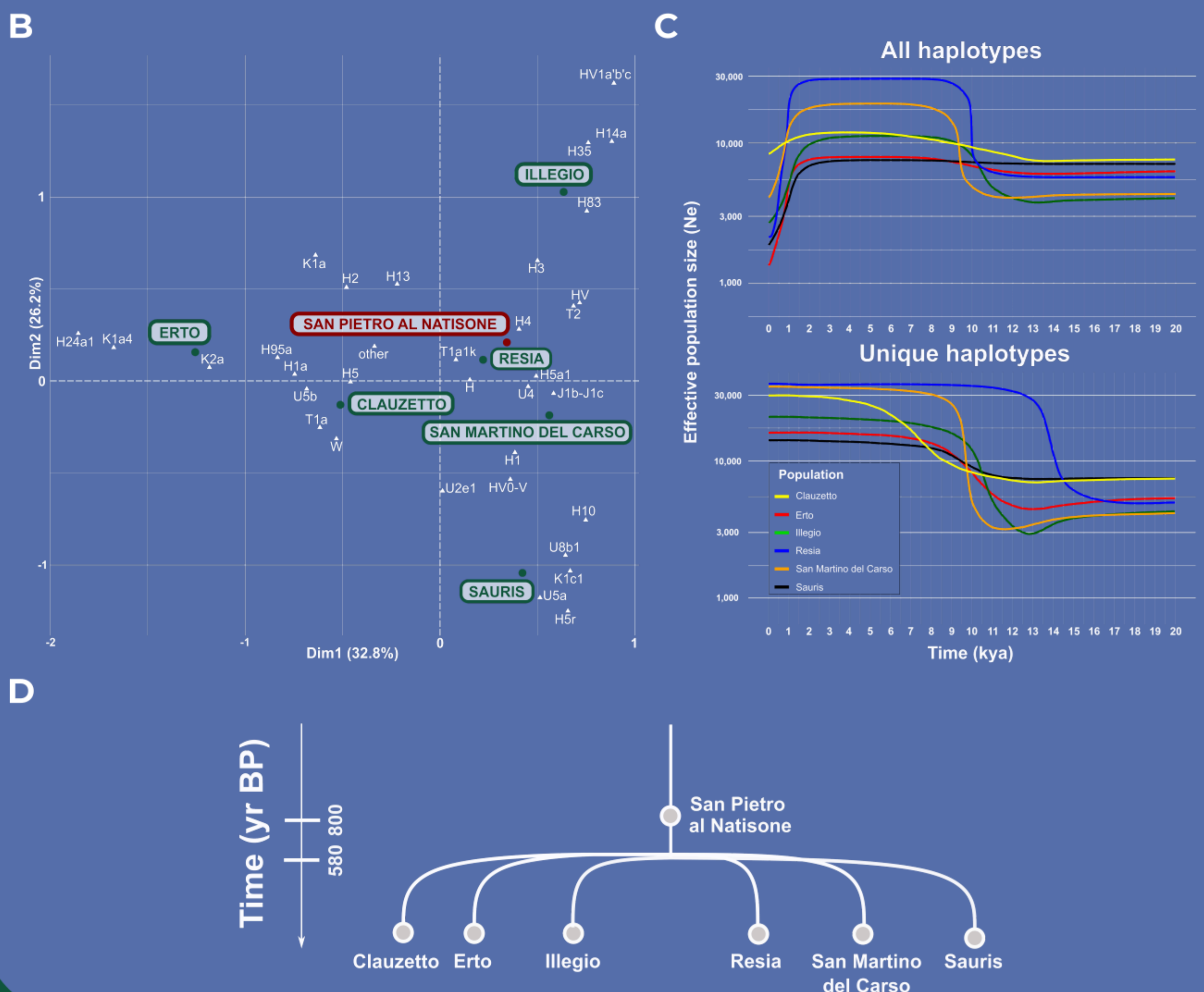
Identify the evolutionary forces that shaped the gene pool of the current inhabitants



**Fig. A)** Geographical framework and haplogroup distribution in Friuli Venezia Giulia  
**Fig. B)** Correspondence Analysis (CA) based on modern and ancient mtDNA haplogroup frequencies.

**Fig. C)** Bayesian Skyline Plots (BSPs) showing changes in effective population size over time for modern mtDNA sequences;

**Fig. D)** ABC inference of demographic history, showing the onset of structure among the six localities from ~580 years ago.



## Bibliography

